Self-checks with explained answers

These are the English self-checks from the course website. The questions come first. Hints, answers and explanations follow all the questions. Attempt a question before reading its feedback. For longer proof and diagram work, use the exercise sheets and their worked solutions.

Lecture 1

1. A functor $F\colon BG \to \mathrm{Vect}_{\mathrm{fd}}$ is the same thing as which of these?
   (a) A representation of G on the space $F(*)$, with $\rho(g) = F(g)$
   (b) An intertwiner between two representations of G
   (c) A linear map $F(*) \to \mathbb{C}$ that every $F(g)$ leaves unchanged
   (d) A functor $\mathrm{Vect}_{\mathrm{fd}} \to BG$ read backwards
2. S_3 permutes the coordinates of $\mathbb{C}^3$, and $\mathbb{C}$ carries the trivial representation. Which linear map $\mathbb{C}^3 \to \mathbb{C}$ is a natural transformation from the first functor $BS_3 \to \mathrm{Vect}_{\mathrm{fd}}$ to the second?
   (a) $x \mapsto x_1 + x_2 + x_3$
   (b) $x \mapsto x_1$
   (c) $x \mapsto x_1 - x_2$
   (d) $x \mapsto x_1 + 2x_2 + 3x_3$
3. Let $V \neq 0$, let $\theta_V\colon V \to V^*$ be a linear isomorphism, and let $f = 2\,\mathrm{id}_V$. Then $f^* \circ \theta_V \circ f = c\,\theta_V$. Find c.
4. Which of these families of maps, one for each finite-dimensional space V, is a natural isomorphism, natural for every linear isomorphism $f\colon V \to W$ (with V^* made covariant by $f \mapsto (f^*)^{-1}$ where it appears)?
   (a) $a_V\colon V \to V^{**}$, $a_V(v)(\varphi) = \varphi(v)$
   (b) $V \to V^*$ sending a chosen basis e_i of V to its dual basis e^i
   (c) On a Hilbert space, the Riesz map $u \mapsto \langle u, \cdot \rangle$, viewed as a map $V \to V^*$
   (d) $V \to V^*$, $v \mapsto 0$
5. How are $\mathrm{Vect}_{\mathrm{fd}}$ and $\mathrm{Mat}_{\mathbb{C}}$ (objects $n = 0, 1, 2, \ldots$, matrices as morphisms) related?
   (a) Equivalent but not isomorphic
   (b) Isomorphic, by $n \mapsto \mathbb{C}^n$
   (c) Isomorphic but not equivalent
   (d) Neither: $\mathrm{Vect}_{\mathrm{fd}}$ has spaces with no basis
6. In the braid groupoid, let a be the braid on three strands that crosses the strands in positions 1 and 2, and b the one that crosses the strands in positions 2 and 3. Forgetting crossings sends $b \circ a$ to a permutation π, with $\pi(i)$ the position where the strand starting at i ends. Find $\pi(1)$.

Lecture 2

1. In Set, how many elements has the coproduct of a set with 2 elements and a set with 3 elements?
2. Let $\xi\colon \mathrm{Hom}(X,-) \Rightarrow \mathrm{Hom}(X',-)$ be a natural isomorphism. Which describes it?
   (a) $\xi_Y(f) = f \circ \varphi$ for all $f\colon X \to Y$, with $\varphi = \xi_X(\mathrm{id}_X)\colon X' \to X$ an isomorphism
   (b) $\xi_Y(f) = \varphi \circ f$ for a fixed isomorphism φ
   (c) $\xi_Y(f) = f \circ \varphi$ for some $\varphi\colon X \to X'$, not necessarily invertible
   (d) Any family of bijections $\mathrm{Hom}(X,Y) \to \mathrm{Hom}(X',Y)$, one for each Y
3. For the adjunction $- \otimes V \dashv \mathrm{Hom}(V,-)$ on $\mathrm{Vect}_{\mathrm{fd}}$, which of these is the counit?
   (a) $g \otimes v \mapsto g(v)$, evaluation
   (b) $u \mapsto (v \mapsto u \otimes v)$
   (c) $g \otimes v \mapsto v$
   (d) $g \otimes v \mapsto g$
4. In $\mathrm{Rep}(S_3)$, let $X = \mathbb{C}^3 \oplus \mathbb{C}^3$, two copies of the permutation representation. What is $\dim \mathrm{End}(X)$?
5. $\mathbb{Z}$ acts on $\mathbb{C}^2$ by $\rho(k) = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$. Why is $\mathbb{C}^2$ not a direct sum of simple objects of $\mathrm{Rep}(\mathbb{Z})$?
   (a) A complement to the invariant line $\mathbb{C}e_1$ would be a second eigenline of $\rho(1)$, which has only one
   (b) Because $\mathbb{Z}$ is abelian, so all its representations are one-dimensional
   (c) Because $\mathbb{C}^2$ has no invariant line at all
   (d) Because $\mathrm{End}(\mathbb{C}^2) = \mathbb{C}\,\mathrm{id}$, so Schur's lemma fails
6. How are $\mathrm{Rep}(S_3)$ and $\mathrm{Rep}(\mathbb{Z}_3)$ related as $\mathbb{C}$-linear categories (forgetting the tensor product)?
   (a) Equivalent: both are semisimple with three simple objects, so both are equivalent to $\mathrm{Vect}_{\mathrm{fd}}^3$
   (b) Not equivalent, because S_3 has a two-dimensional irreducible representation
   (c) Not equivalent, because S_3 is not abelian
   (d) Equivalent, because S_3 and $\mathbb{Z}_3$ have the same number of elements

Lecture 3

1. In Vec_G^ω on $\mathbb{Z}_2$ with ω normalised and $\omega(s,s,s) = -1$, the objects $(s\otimes s)\otimes s$ and $s\otimes(s\otimes s)$ are both s, yet $\alpha_{s,s,s} = -\mathrm{id}_s \neq \mathrm{id}_s$. Why does this not contradict coherence?
   (a) Coherence speaks only of formal morphisms, and id_s is not a formal morphism between the two bracketings
   (b) Coherence holds only for strict monoidal categories
   (c) The pentagon fails in this category, so it is not monoidal
   (d) The associator of Vec_G^ω is not natural
2. Which equation is the interchange law, for composable f, g and composable h, k?
   (a) $(g\otimes k)\circ(f\otimes h) = (g\circ f)\otimes(k\circ h)$
   (b) $(g\otimes k)\circ(f\otimes h) = (g\circ h)\otimes(k\circ f)$
   (c) $(f\otimes h)\circ(g\otimes k) = (g\circ f)\otimes(k\circ h)$
   (d) $f\otimes h = h\otimes f$
3. In a general monoidal category, which statement about $s, t \in \mathrm{End}(1)$ is always true?
   (a) $s\circ t = t\circ s$
   (b) $s\otimes \mathrm{id}_X = \mathrm{id}_X \otimes s$ for every object X
   (c) $\mathrm{End}(X)$ is commutative for every object X
   (d) s is a complex number
4. In Vec_G^ω on $\mathbb{Z}_2 = \{1, s\}$ with ω normalised, the pentagon at (s,s,s,s) reads $1 = \omega(s,s,s)^n$. Find n.
5. In Vec_G^ω the unitors are identities and $\alpha_{g,h,k} = \omega(g,h,k)\ \mathrm{id}_{ghk}$ on simple objects. What does the triangle axiom require?
   (a) $\omega(g,1,h) = 1$ for all g, h
   (b) $\omega(g,g^{-1},g) = 1$ for all g
   (c) $\omega(g,h,k) = \omega(k,h,g)$ for all g, h, k
   (d) Nothing: with identity unitors the triangle holds automatically
6. A monoidal functor $\mathrm{Vec}_G^\omega \to \mathrm{Vec}_G^{\omega'}$ on $\mathbb{Z}_2$ is the identity on objects, with $J_{g,h} = j(g,h)\ \mathrm{id}$ and $J_0 = j_0\ \mathrm{id}$, and both cocycles are normalised. What must hold?
   (a) $\omega'(s,s,s) = \omega(s,s,s)$
   (b) Nothing: $j(s,s)$ can always be chosen to make ω' anything
   (c) $\omega'(s,s,s) = -\omega(s,s,s)$
   (d) $\omega'(s,s,s) = j(s,s)^2\,\omega(s,s,s)$

Lecture 4

1. In $\mathrm{Vect}_{\mathrm{fd}}$, with the duals of Lecture 4, what is the rotated box $f^* : Y^* \to X^*$ of a linear map $f : X \to Y$?
   (a) $\varphi \mapsto \varphi \circ f$, the transpose
   (b) The adjoint $f^\dagger$ for a chosen inner product
   (c) $(f^{-1})^*$, the inverse transpose
   (d) $\varphi \mapsto f \circ \varphi$
2. Let X and Y have left duals X^* and Y^* in a monoidal category. Which object, with a nested cup and cap, is a left dual of $X \otimes Y$?
   (a) $Y^* \otimes X^*$
   (b) $X^* \otimes Y^*$
   (c) $X \otimes Y$ itself
   (d) None, unless the category has a braiding
3. Teleportation with the normalised Bell vector $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$: the map $|\psi\rangle \mapsto (\langle\Phi^+| \otimes \mathrm{id})(|\psi\rangle \otimes |\Phi^+\rangle)$ is c id on a qubit. Find c.
4. On Vec_G for $G = \mathbb{Z}_2 = \{1, s\}$, take the pivotal structure $a_s = -\mathrm{id}$ of Lecture 4. Find the quantum dimension $d_{s\otimes s\otimes s}$.
5. For spin-$\frac{1}{2}$, $V = \mathbb{C}^2$, the cap $-\Omega^\dagger$ makes both snakes with the singlet cup Ω straight, and closes the loop to $-\Omega^\dagger\Omega = -2$. With the pivotal structure a of $\mathrm{Vect}_{\mathrm{fd}}$, what is the quantum dimension d_V?
   (a) 2
   (b) -2
   (c) It depends on which cap is paired with Ω
   (d) It is not defined, since V is identified with its dual
6. With the canonical pivotal structure on representations, which of these objects has Frobenius–Schur indicator $\kappa = -1$?
   (a) Spin-$\frac{1}{2}$ of SU(2)
   (b) The standard representation of S_3
   (c) Spin-1 of SU(2)
   (d) The sign representation of S_3

Lecture 5

1. In $\mathrm{SU}(2)_4$, with simple objects $j = 0, \frac{1}{2}, \ldots, 2$, what is $1 \otimes 2$?
   (a) 1
   (b) $1 \oplus 2 \oplus 3$
   (c) $1 \oplus 2$
   (d) $0 \oplus 1 \oplus 2$
2. Which identity holds for the multiplicities N_{ab}^{c} of every fusion category?
   (a) $N_{ab}^{c} = N_{bc^*}^{a^*}$
   (b) $N_{ab}^{c} = N_{ba}^{c}$
   (c) $N_{ab}^{c} = N_{a^*b^*}^{c^*}$
   (d) $N_{ab}^{1} = 1$ for all a and b
3. Find the global dimension $\dim C = \sum_a d_a^2$ of the Fibonacci category, with its canonical spherical structure.
4. How many states do five Fibonacci anyons τ have with total charge τ? That is, find $\dim V_{\tau}^{\tau\tau\tau\tau\tau}$.
5. In the isotopy-invariant normalisation of Lecture 5, what is the bubble $\psi_c^{ab\dagger} \circ \psi_c^{ab}$, for $N_{ab}^{c} = 1$?
   (a) $\sqrt{d_a d_b / d_c}\ \mathrm{id}_c$
   (b) id_c
   (c) $\sqrt{d_c / (d_a d_b)}\ \mathrm{id}_c$
   (d) $d_c\ \mathrm{id}_c$
6. On Vec_G for $G = \mathbb{Z}_2 = \{1, s\}$, take the pivotal structure $a_s = -\mathrm{id}$ of Lecture 4. What are $\mathrm{FPdim}(s)$ and d_s?
   (a) $\mathrm{FPdim}(s) = 1$ and $d_s = -1$
   (b) Both are -1
   (c) Both are 1
   (d) $\mathrm{FPdim}(s) = -1$ and $d_s = 1$

Lecture 6

1. In a unitary multiplicity-free fusion category with the vertices of Lecture 5, why is every F-matrix F_d^{abc} unitary?
   (a) The left and right trees are orthogonal bases with one common norm, and the associator is unitary
   (b) The pentagon equation forces every solution to be unitary
   (c) Any invertible matrix of F-symbols becomes unitary after a suitable gauge transformation
   (d) F-matrices are permutation matrices up to phases
2. In Ising, with $F_{\sigma}^{\sigma\sigma\sigma} = \kappa H$ and $\kappa = \pm 1$, take the pentagon equation of Lecture 6 at $a = b = c = d = \sigma$, $e = \psi$, $f = 1$, $g = k = \sigma$ and $l = 1$. Find $y = [F_{\psi}^{\sigma\psi\sigma}]_{\sigma\sigma}$.

3. With the gauge law of Lecture 6, which of these F-symbols is gauge invariant?
   (a) $[F_\sigma^{\sigma\sigma\sigma}]_{11}$ in Ising
   (b) $[F_\sigma^{\sigma\sigma\sigma}]_{1\psi}$ in Ising
   (c) $[F_\tau^{\tau\tau\tau}]_{1\tau}$ in Fibonacci
   (d) $F_\psi^{\sigma\psi\sigma}$ in Ising
4. In the unitary Fibonacci category, find $[F_\tau^{\tau\tau\tau}]_{11}$, the coefficient of the right tree with $f = 1$ in the F-move of the left tree with $e = 1$.
5. With the fusion rule $\tau \otimes \tau \cong 1 \oplus \tau$, how many solutions does the pentagon have up to gauge transformations with $u \neq 0$ (unitary or not)?
6. Ising's fusion rules carry the solutions $F_\sigma^{\sigma\sigma\sigma} = H$ and $F_\sigma^{\sigma\sigma\sigma} = -H$ (with the other F-symbols of Lecture 6). How are they related?
   (a) They are inequivalent: $\kappa = d_\sigma [F_\sigma^{\sigma\sigma\sigma}]_{11} = \pm 1$ is a gauge invariant
   (b) They are equivalent, since $-H$ and H agree up to a global phase
   (c) A gauge with $u_1^{\sigma\sigma} = -1$ maps one to the other
   (d) Their fusion rules differ

Lecture 7

1. Which relation of the symmetric group S_n, in its generators σ_i, fails in the braid group B_n?
   (a) $\sigma_i^2 = 1$
   (b) $\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}$
   (c) $\sigma_i \sigma_j = \sigma_j \sigma_i$ for $|i - j| \geq 2$
   (d) None: B_n and S_n are isomorphic
2. A constant matrix R on $V \otimes V$ satisfies $R_{12} R_{13} R_{23} = R_{23} R_{13} R_{12}$ exactly when which matrix satisfies the braid form $c_{12} c_{23} c_{12} = c_{23} c_{12} c_{23}$?
   (a) $c = P \circ R$, with P the flip
   (b) $c = R$
   (c) $c = R^{-1}$
   (d) $c = R^T$
3. Two instances of Fibonacci's hexagon equations give $R_\tau^2 = R_1$ and $R_\tau^2 + \varphi^{-1} R_\tau + 1 = 0$, for $R_1 = R_1^{\tau\tau}$ and $R_\tau = R_\tau^{\tau\tau}$. Take the conjugate chirality, $R_1 = e^{4\pi i/5}$. Then $R_\tau = e^{2\pi i s}$ with $0 \leq s < 1$. Find s.
4. With $R_1^{\tau\tau} = e^{-4\pi i/5}$, $R_\tau^{\tau\tau} = e^{3\pi i/5}$ and $d_\tau = \varphi$, the topological spin is $\theta_\tau = e^{2\pi i s}$ with $0 \leq s < 1$. Find s.
5. How many braidings, in total, do the two categories Vec_G^ω on $\mathbb{Z}_2$ (with $\omega(s,s,s) = 1$ and with $\omega(s,s,s) = -1$) carry?
6. Which of these is unchanged by every gauge transformation of the vertices?
   (a) The double braiding $R_c^{ab} R_c^{ba}$
   (b) R_c^{ab} for $a \neq b$, neither of them 1
   (c) $(R_c^{ab})^2$ for $a \neq b$, neither of them 1
   (d) $[F_\sigma^{\sigma\sigma\sigma}]_{1\psi}$ in Ising

Lecture 8

1. For the enveloping algebra $U(\mathfrak{sl}_2)$, which coproduct on $x \in \mathfrak{sl}_2$ makes $\mathrm{Rep}(U(\mathfrak{sl}_2))$ monoidal, with x acting on $V \otimes W$ by $\Delta(x)$ and $\mathbb{C}$ as the unit object?
   (a) $\Delta(x) = x \otimes 1 + 1 \otimes x$
   (b) $\Delta(x) = x \otimes x$
   (c) $\Delta(x) = x \otimes 1$
   (d) $\Delta(x) = \frac{1}{2}(x \otimes 1 + 1 \otimes x)$
2. In $\mathrm{Rep}(H)$ the dual V^* has $(h \cdot f)(v) = f(S_H(h)v)$. For $U(\mathfrak{sl}_2)$, with $S_H(x) = -x$ and x acting on V by the matrix $\rho(x)$, which matrix gives the action of x on V^*, acting on the column of coefficients of f in the dual basis?
   (a) $-\rho(x)^T$
   (b) $\rho(x)^T$
   (c) $-\rho(x)$
   (d) $\rho(x)^{-1}$
3. On $\mathbb{C}[\mathbb{Z}_2] = \mathrm{span}(1, g)$ take $R = \frac{1}{2}(1 \otimes 1 + 1 \otimes g + g \otimes 1 - g \otimes g)$, which acts on $V_a \otimes V_b$ by $(-1)^{ab}$, where V_a is the one-dimensional module on which g acts by $(-1)^a$, for $a = 0, 1$. Let $W = V_0 \oplus V_1$. Find the trace of the braiding $c_{W,W}$ on $W \otimes W$.
4. Which of these fusion categories is equivalent to $\mathrm{Rep}(H)$ for a finite-dimensional Hopf algebra H?
   (a) Vec_G for $G = \mathbb{Z}_3$, whatever its pivotal structure
   (b) Fibonacci
   (c) Ising
   (d) Vec_G^ω on $\mathbb{Z}_2$ with $\omega(s, s, s) = -1$
5. On $V_1 \otimes V_1$, B_q has eigenvalue q on the triplet and $-q^{-1}$ on the q-singlet. Find $\det B_q$ at $q = 2$.
6. B_q satisfies the braid relation, but for the q-singlet cup v it gives $B_1 B_2 (v \otimes \mathrm{id}) = q\,(\mathrm{id} \otimes v)$, while a braiding must give $c_1 c_2 (v \otimes \mathrm{id}) = \mathrm{id} \otimes v$. Which rescaling $c = \lambda B_q$ is a braiding?
   (a) $\lambda = A^{-1}$, for a root A of $q = A^2$
   (b) $\lambda = 1$: B_q is already a braiding
   (c) $\lambda = q^{-1}$
   (d) $\lambda = A$, for a root A of $q = A^2$

Lecture 9

1. In a ribbon category, which formula gives the twist of a tensor product?
   (a) $\theta_{X\otimes Y} = c_{Y,X}\,c_{X,Y}\,(\theta_X \otimes \theta_Y)$
   (b) $\theta_{X\otimes Y} = \theta_X \otimes \theta_Y$
   (c) $\theta_{X\otimes Y} = c_{X,Y}\,(\theta_X \otimes \theta_Y)$
   (d) $\theta_{X\otimes Y} = (c_{Y,X}\,c_{X,Y})^{-1}(\theta_X \otimes \theta_Y)$
2. With the skein relation (crossing $= A\cdot \mathrm{id} + A^{-1}\cdot$ cup$\circ$cap) and the loop $\delta = -A^2 - A^{-2}$, a positive curl on one strand is λ times the strand. At $A = e^{i\pi/6}$, $\lambda = e^{2\pi i s}$ with $0 \le s < 1$. Find s.
3. Which identity makes the Kauffman bracket invariant under Reidemeister II?
   (a) $\delta = -A^2 - A^{-2}$
   (b) $\delta = -2$
   (c) $\delta = A^2 + A^{-2}$
   (d) $A^4 = 1$
4. Close both strands of $p_2 = 1 - U/\delta$ to compute its trace, at $A = e^{i\pi/12}$, the level $k = 4$.
5. At $A = e^{i\pi/(2(k+2))}$, why is the projector p_{k+1} discarded in building $\mathrm{SU}(2)_k$?
   (a) It is negligible: its trace $(-1)^{k+1}[k+2]_q$ vanishes, and it closes to zero in every network
   (b) The recursion cannot define it, because $[k+1]_q = 0$
   (c) It is not idempotent at a root of unity
   (d) Its twist is 1, so it is the vacuum
6. In the Temperley–Lieb pivotal structure the coloured strand n has twist $\theta_n = (-1)^n A^{n(n+2)}$. At $k = 2$, $A = e^{i\pi/8}$, spin $\frac{1}{2}$ is $n = 1$. After the pivotal flip, its twist is $e^{2\pi i s}$ with $0 \le s < 1$. Find s.

Lecture 10

1. In the toric code, with $\theta = (1, 1, 1, -1)$ on $(1, e, m, \varepsilon)$ and $\dim C = 4$, find the entry S_{em} of the normalised S-matrix.
2. Which of these premodular categories is modular?
   (a) Fibonacci
   (b) sVec
   (c) $\mathrm{Rep}(S_3)$ with its symmetric braiding
   (d) Vec_G on $\mathbb{Z}_2$ with the trivial braiding $c_{s,s} = 1$
3. Ising's S-matrix in the order $1, \sigma, \psi$ is $\frac{1}{2}\begin{pmatrix} 1 & \sqrt{2} & 1 \\ \sqrt{2} & 0 & -\sqrt{2} \\ 1 & -\sqrt{2} & 1 \end{pmatrix}$. What does the Verlinde formula give for $\sigma \otimes \sigma$?
   (a) $1 \oplus \psi$
   (b) $1 \oplus \sigma \oplus \psi$
   (c) σ
   (d) $1 \oplus \sigma$
4. The semion has $d_s = 1$ and $\theta_s = i$. From the Gauss sum, find its central charge c, modulo 8, in $[0, 8)$.

5. In Fibonacci, a loop labelled τ around a strand τ (as in Lecture 10's loop picture) is λ times the strand. Find λ.
6. The killing loop $\omega_0 = (\dim C)^{-1} \sum_b d_b\, b$ around a simple strand x is $\delta_{x,1}\,\mathrm{id}_x$. In the proof, which hypotheses rule out a nonzero killing loop around a simple $x \neq 1$?
   (a) Only 1 is transparent, and every $d_c > 0$ and $|\theta_c| = 1$ (unitarity)
   (b) Every d_b is an integer
   (c) The category is pointed
   (d) The S-matrix is symmetric

Lecture 11

1. In Vec_G^ω, the twisted group algebra $\mathbb{C}_\psi[H]$ has $\mu = \psi(g,h)$ on $\delta_g \otimes \delta_h \to \delta_{gh}$ and the associator $\alpha_{g,h,k} = \omega(g,h,k)$. Which equation is its associativity?
   (a) $\psi(g,h)\,\psi(gh,k) = \omega(g,h,k)\,\psi(h,k)\,\psi(g,hk)$
   (b) $\psi(g,h)\,\psi(gh,k) = \psi(h,k)\,\psi(g,hk)$
   (c) $\psi(g,h)\,\psi(gh,k)\,\omega(g,h,k) = \psi(h,k)\,\psi(g,hk)$
   (d) $\psi(g,h)\,\psi(h,k) = \omega(g,h,k)\,\psi(gh,k)\,\psi(g,hk)$
2. How many indecomposable semisimple module categories does Vec_G^ω have, for $G = \mathbb{Z}_2$ and $\omega(s,s,s) = -1$?
3. How many indecomposable semisimple module categories does Vec_G have for $G = \mathbb{Z}_4$ (with $\omega = 1$)? Every 2-cocycle on a cyclic group is a coboundary.
4. For the regular module of Vec_G, with G non-abelian, what is the internal Hom $[\delta_x, \delta_y]$?
   (a) $\delta_{yx^{-1}}$
   (b) $\delta_{x^{-1}y}$
   (c) $\delta_{xy^{-1}}$
   (d) $\delta_{y^{-1}x}$
5. Which of these statements is true?
   (a) Every indecomposable semisimple module category over a fusion category C is equivalent to $\mathrm{Mod}_C(A)$ for an algebra A in C
   (b) Every indecomposable module category over C is the regular one, $\mathrm{Mod}_C(1)$
   (c) $\mathrm{Vect}_{\mathrm{fd}}$ is a module category over every Vec_G^ω, with G acting trivially
   (d) The internal End $[M,M]$ of an object of a module category is always the unit 1
6. In Vec_G with $G = \mathbb{Z}_6$, take the algebra $A = \mathbb{C}[H]$ for $H = \{0,2,4\} \cong \mathbb{Z}_3$. How many simple objects does the module category $\mathrm{Mod}(A)$ have?

Lecture 12

1. For the regular module of Vec_G, with G non-abelian, the dual $(\mathrm{Vec}_G)^*_{\mathrm{Vec}_G}$ has tensor product $F \otimes F' = F \circ F'$. Which assignment is a tensor functor $\mathrm{Vec}_G \to (\mathrm{Vec}_G)^*_{\mathrm{Vec}_G}$?
   (a) $g \mapsto (- \otimes \delta_{g^{-1}})$
   (b) $g \mapsto (- \otimes \delta_g)$
   (c) $g \mapsto (\delta_g \otimes -)$
   (d) $g \mapsto (\delta_{g^{-1}} \otimes -)$
2. In $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$ the simple object (g, χ) carries the half-braiding $\gamma_{\delta_h} = \chi(h)\,\mathrm{id}$ on $\delta_h \otimes \delta_g$, with $e = (0, \chi_-)$ and $m = (1, 1)$. What are the braidings $c_{e,m}$ and $c_{m,e}$ of the centre?
   (a) $c_{e,m} = 1$ and $c_{m,e} = -1$
   (b) $c_{e,m} = -1$ and $c_{m,e} = 1$
   (c) $c_{e,m} = c_{m,e} = -1$
   (d) $c_{e,m} = c_{m,e} = 1$
3. In Vec_{S_3}, let g be a transposition. How many half-braidings does the simple object δ_g carry, on its own?
4. How many simple objects does $\mathcal{Z}(\mathrm{Vec}_G)$ have for the dihedral group $G = D_4$ of order 8? Its classes are $\{1\}$, $\{r^2\}$, $\{r, r^3\}$, $\{s, sr^2\}$ and $\{sr, sr^3\}$.
5. Find $\dim \mathcal{Z}(\mathrm{Fib})$, where $\dim \mathrm{Fib} = 1 + \varphi^2$ and $\varphi = (1 + \sqrt{5})/2$. Give three decimal places.
6. Which of these statements is true?
   (a) Vec_G and $\mathrm{Rep}(G)$ are Morita equivalent for every finite group G
   (b) $\mathrm{Vec}_{\mathbb{Z}_2}$ and its twist by $\omega(s, s, s) = -1$ are Morita equivalent, since their fusion rules agree
   (c) Morita equivalent fusion categories are equivalent as tensor categories
   (d) Fib and $\mathrm{Vec}_{\mathbb{Z}_2}$ are Morita equivalent, since each has two simple objects

Lecture 13

1. In $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, for which non-trivial simple X is there no Lagrangian algebra on $1 \oplus X$?
   (a) $X = \varepsilon$
   (b) $X = e$
   (c) $X = m$
2. How many gapped boundaries does the bulk $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2 \times \mathbb{Z}_2})$ have, counting them as its Lagrangian algebras? Use $H^2(\mathbb{Z}_2 \times \mathbb{Z}_2, \mathbb{C}^\times) = \mathbb{Z}_2$, and $H^2 = 0$ for cyclic groups.
3. At the smooth boundary of the toric code, where the lattice ends along a row of edges, which anyon condenses?
   (a) m
   (b) e
   (c) ε
   (d) None of them

4. On the periodic transverse-field Ising chain, with the translation $\mathcal{T} O_j \mathcal{T}^{-1} = O_{j+1}$, what is the square of the Kramers–Wannier operator D?
   (a) $D^2 = (1+\eta)\,\mathcal{T}^{-1}$
   (b) $D^2 = 1+\eta$
   (c) $D^2 = 1$
   (d) $D^2 = (1+\eta)\,\mathcal{T}$
5. Find the global dimension $\sum_X d_X^2$ of the Tambara–Yamagami category
$$\mathrm{TY}(\mathbb{Z}_3, \chi, \pm 1/\sqrt{3}).$$
6. Which of these fusion categories cannot be fully gauged, because it has no module category with one simple object?
   (a) Vec_G^{ω} for $G = \mathbb{Z}_2$ and $\omega(s,s,s) = -1$
   (b) $\mathrm{Vec}_{\mathbb{Z}_2}$
   (c) Vec_{S_3}
   (d) $\mathrm{Vec}_{\mathbb{Z}_2 \times \mathbb{Z}_2}$

Lecture 14

1. Why is the Frobenius algebra $A = Z(S^1)$ of a 2D TQFT commutative?
   (a) The swap of the two legs of the pair of pants is a bordism, and a symmetric monoidal Z sends it to the flip of Vect
   (b) Every Frobenius algebra is commutative
   (c) The pair of pants is drawn left–right symmetric, so μ is
   (d) Vect is braided, and a braided target makes every algebra commutative
2. Let $A = \mathbb{C}^2$ with orthogonal idempotents e_1, e_2 and counit $\varepsilon_A(e_1) = 1$, $\varepsilon_A(e_2) = \frac{1}{2}$. Find the partition function $Z(\Sigma_3)$ of its 2D TQFT on the closed surface of genus 3.
3. The dihedral group D_4 of order 8 has irreducible representations of dimensions $1, 1, 1, 1, 2$. Find the 2D Dijkgraaf–Witten partition function $Z(\Sigma_2)$, normalised as $\#\,\mathrm{Hom}(\pi_1 \Sigma_2, G)/|G|$.
4. For the Turaev–Viro theory of $C = \mathrm{Vec}_{S_3}$, find the partition function $Z_{\mathrm{TV},C}(T^2 \times S^1)$ of the closed 3-manifold $T^2 \times S^1$.
5. Which of these unitary modular categories gives a Reshetikhin–Turaev theory without a framing anomaly?
   (a) The toric code $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, with $c = 0$
   (b) Fibonacci, with $c = \frac{14}{5}$
   (c) The semion, with $c = 1$
   (d) Ising, with $c = \frac{1}{2}$
6. What input does the Turaev–Viro state sum need?
   (a) A spherical fusion category with $\dim C \neq 0$; no braiding is used
   (b) A braided fusion category, whose braiding weights the tetrahedra
   (c) Any fusion category, with or without a spherical structure

Hints, answers and explanations

Lecture 1

1. **Hint.** BG has one object $*$ and one morphism for each $g \in G$, composed by multiplication. What does a functor do to that object and to those morphisms?
   Answer. A representation of G on the space $F(*)$, with $\rho(g) = F(g)$
   F gives a space $V = F(*)$ and maps $F(g)$ with $F(gh) = F(g)F(h)$ and $F(1) = \mathrm{id}_V$; they are invertible because functors keep isomorphisms. That is a representation. An intertwiner is a natural transformation between two such functors, one level up.
2. **Hint.** With one object, naturality is one equation for each g: the map must not notice that the coordinates were permuted.
   Answer. $x \mapsto x_1 + x_2 + x_3$
   Naturality reads $T(\rho(\pi)x) = T(x)$ for every permutation π, so T must be symmetric in the coordinates. Only the sum is. The coordinate x_1 fails at the transposition $(1\,2)$: it sends e_1 to 1 but $\rho((1\,2))e_1 = e_2$ to 0.
3. **Hint.** The dual map of $\lambda\ \mathrm{id}_V$ is $\lambda\ \mathrm{id}_{V^*}$, and θ_V is linear.
   Answer. 4
   $f^*\theta_V f = 2 \cdot 2\,\theta_V = 4\theta_V$. Naturality would need $f^*\theta_V f = \theta_V$, so $3\theta_V = 0$: no natural family of linear isomorphisms $V \to V^*$ survives this test. The lecture uses $f = i\ \mathrm{id}_V$, which gives $-\theta_V$ and is unitary as well.
4. **Hint.** Test each candidate with $f = i\ \mathrm{id}_V$, and remember that a natural isomorphism needs every component to be invertible.
   Answer. $a_V : V \to V^{**}$, $a_V(v)(\varphi) = \varphi(v)$
   $f^{**}(a_V(v))(\psi) = \psi(f(v)) = a_W(f(v))(\psi)$, so a is natural, and each a_V is invertible in finite dimension. A linear $\theta_V : V \to V^*$ natural for $f = i\ \mathrm{id}_V$ would satisfy $-\theta_V = \theta_V$, so it is zero: the basis map fails, and the zero map is natural but not an isomorphism. The Riesz map passes the test only because it is antilinear, $R(iu) = -iR(u)$, so it is not a linear map at all.
5. **Hint.** An isomorphism of categories is injective on objects and keeps isomorphic objects isomorphic. How many objects of $\mathrm{Mat}_{\mathbb{C}}$ are isomorphic to 1?
   Answer. Equivalent but not isomorphic
   Choosing a basis of every space gives a functor back and a natural isomorphism, so they are equivalent. They are not isomorphic: an isomorphism of categories is injective on objects and keeps isomorphic objects isomorphic, but the line $\{(z, 0)\} \subset \mathbb{C}^2$ and $\mathbb{C}$ are different, isomorphic objects of $\mathrm{Vect}_{\mathrm{fd}}$, while in the skeletal $\mathrm{Mat}_{\mathbb{C}}$ different objects are never isomorphic. Every isomorphism of categories is an equivalence, so "isomorphic but not equivalent" cannot happen.
6. **Hint.** In $b \circ a$, which braid acts first, and is it drawn at the bottom or at the top?
   Answer. 3
   In $b \circ a$ the braid a acts first, so it is the lower layer: time runs upward. The strand starting at 1 goes to position 2 in a, then to 3 in b, so $\pi(1) = 3$. Reading the picture downward would apply b first and give 2.

Lecture 2

1. **Hint.** The coproduct is defined by maps out of it: a pair of maps $g_k \colon X_k \to Z$ must factor through exactly one map. Which construction does that?
   Answer. 5
   The coproduct in Set is the disjoint union, with $2 + 3 = 5$ elements. The product is $X_1 \times X_2$, with 6; in Set the two differ, while in Vect the direct sum is both.
2. **Hint.** Apply naturality to the morphism $f \colon X \to Y$ and the element id_X of $\mathrm{Hom}(X, X)$.
   Answer. $\xi_Y(f) = f \circ \varphi$ for all $f \colon X \to Y$, with $\varphi = \xi_X(\mathrm{id}_X) \colon X' \to X$ an isomorphism
   Naturality gives $\xi_Y(f) = \xi_Y(f \circ \mathrm{id}_X) = f \circ \xi_X(\mathrm{id}_X)$, so ξ is precomposition with $\varphi = \xi_X(\mathrm{id}_X) \colon X' \to X$, and the inverse transformation gives its inverse. Postcomposing $f \colon X \to Y$ with a fixed φ does not even give a morphism $X' \to Y$, since φ would have to start at Y for every Y; and an arbitrary family of bijections need not be natural at all.
3. **Hint.** The counit is the map $F(G(W)) \to W$ that corresponds to $\mathrm{id}_{G(W)}$ under the bijection.
   Answer. $g \otimes v \mapsto g(v)$, evaluation
   Under Θ, the identity of $\mathrm{Hom}(V, W)$ corresponds to the map $g \otimes v \mapsto g(v)$, evaluation. The map $u \mapsto (v \mapsto u \otimes v)$ is the unit, which goes the other way. The rules $g \otimes v \mapsto v$ and $g \otimes v \mapsto g$ are not bilinear in (g, v), so they do not even define linear maps on the tensor product.
4. **Hint.** First decompose $\mathbb{C}^3$ into simple objects; then use
   $$\dim \mathrm{Hom}(X, Y) = \sum_i m_i(X)\, m_i(Y).$$
   Answer. 8
   $\mathbb{C}^3 \cong \mathbf{1} \oplus \mathrm{std}$, so $X \cong \mathbf{1}^{\oplus 2} \oplus \mathrm{std}^{\oplus 2}$, with multiplicities 2, 0 and 2, and $\dim \mathrm{End}(X) = 2^2 + 0^2 + 2^2 = 8$: $\mathrm{End}(X) \cong M_2(\mathbb{C}) \times M_2(\mathbb{C})$. Squaring $\dim X = 6$, or adding the multiplicities, gives the wrong answer.
5. **Hint.** A direct sum of two invariant lines would give two independent eigenvectors of $\rho(1)$. How many does it have?
   Answer. A complement to the invariant line $\mathbb{C}e_1$ would be a second eigenline of $\rho(1)$, which has only one
   $\rho(1)$ has the single eigenvalue 1 with the single eigenline $\mathbb{C}e_1$, so $\mathbb{C}e_1$ has no invariant complement, and $\mathbb{C}^2$ is indecomposable but not simple. $\mathbb{C}e_1$ is an invariant line, abelian groups can have indecomposable higher-dimensional representations like this one, and $\mathrm{End}(\mathbb{C}^2)$ contains $\rho(1) - I$, so it is bigger than $\mathbb{C}$ id.
6. **Hint.** What does the categorical Schur theorem say a semisimple category is determined by, up to equivalence?
   Answer. Equivalent: both are semisimple with three simple objects, so both are equivalent to $\mathrm{Vect}_{\mathrm{fd}}^3$
   Up to equivalence a semisimple category is determined by its number of simple objects, and both have three (one per conjugacy class). The dimensions of the representations and whether the group is abelian are invisible to the linear category; the tensor product of Lecture 3 tells them apart. And S_3 has 6 elements, $\mathbb{Z}_3$ has 3.

Lecture 3

1. **Hint.** First check whether Vec_G^ω is monoidal at all, so that coherence applies. Then: of which morphisms, between which objects, does coherence speak?
   Answer. Coherence speaks only of formal morphisms, and id_s is not a formal morphism between the two bracketings
   As formal words, $(x \otimes x) \otimes x$ and $x \otimes (x \otimes x)$ are different, and no formal morphism between them is an identity; id_s compares them only because they happen to be the same object. The category is monoidal (ω is normalised, which is the triangle, and the pentagon holds since $\omega(s,s,s)^2 = 1$), coherence applies to every monoidal category, and the associator is natural because morphisms between simple objects are scalars.
2. **Hint.** Draw both sides: boxes in two columns, the first factor on the left, time upward. Read the picture row by row and column by column.
   Answer. $(g \otimes k) \circ (f \otimes h) = (g \circ f) \otimes (k \circ h)$
   Row by row the picture is $(g \otimes k) \circ (f \otimes h)$; column by column it is $(g \circ f) \otimes (k \circ h)$, with f below g on the left strand and h below k on the right. Mixing the strands, or composing in the wrong order, is not even defined in general, since the sources and targets do not match; and $f \otimes h$ and $h \otimes f$ do not have the same source.
3. **Hint.** Test each statement in Vect first. Then ask which of those that survive the lecture proves for every monoidal category.
   Answer. $s \circ t = t \circ s$
   Eckmann–Hilton: in a strict category $s \circ t = s \otimes t = t \circ s$ by interchange, and so in every monoidal category by coherence. Moving a scalar across a strand is not a planar deformation, and $s \otimes \mathrm{id}_X$ and $\mathrm{id}_X \otimes s$ can differ. $\mathrm{End}(X)$ is not commutative in general (take $X = \mathbb{C}^2$ in Vect), and $\mathrm{End}(1)$ need not be $\mathbb{C}$.
4. **Hint.** The two-arrow path is $\omega(s^2,s,s)\,\omega(s,s,s^2)$ and the three-arrow path is $\omega(s,s,s)\,\omega(s,s^2,s)\,\omega(s,s,s)$. Use $s^2 = 1$ and normalisation.
   Answer. 2
   With $s^2 = 1$ and $\omega = 1$ whenever an argument is 1, the two-arrow path is 1 and the three-arrow path is $\omega(s,s,s)^2$, so $\omega(s,s,s) = \pm 1$: two monoidal categories with the same fusion rules.
5. **Hint.** Take $X = g$ and $Y = h$ in the triangle. Each arrow is a scalar times an identity; compute the scalar along each path.
   Answer. $\omega(g,1,h) = 1$ for all g, h
   Along $\alpha_{g,1,h}$ followed by $\mathrm{id} \otimes l_h$ the scalar is $\omega(g,1,h)$; along $r_g \otimes \mathrm{id}$ it is 1. So the triangle is exactly normalisation, and the identity unitors do not make it automatic, because the associator need not be the identity. On $\mathbb{Z}_2$ the cocycle with $\omega(s,s,s) = -1$ gives a monoidal category, so $\omega(g,g^{-1},g) = 1$ is not required; and for $\mathbb{Z}_3 = \{1, r, r^2\}$ with $p = 1$, $\omega(r,r,r^2) = e^{2\pi i/3}$ while $\omega(r^2,r,r) = 1$, so no symmetry of the arguments is required either.
6. **Hint.** Write the associativity condition at (s,s,s) and see which j's cancel; then use the unit conditions.
   Answer. $\omega'(s,s,s) = \omega(s,s,s)$
   At (s,s,s) the condition reads $\omega'(s,s,s)\,j(s,s)\,j(s,1) = j(s,s)\,j(1,s)\,\omega(s,s,s)$; $j(s,s)$ cancels, and the unit conditions give $j(s,1) = j(1,s)$. So $\omega(s,s,s)$ cannot be changed this way, and the two $\mathbb{Z}_2$ categories are inequivalent (with EGNO's classification).

Lecture 4

1. **Hint.** Draw f^*: a cap, the box, a cup. Feed it a functional φ on Y and evaluate the result on a vector $v \in X$.
   Answer. $\varphi \mapsto \varphi \circ f$, the transpose
   Following the picture, φ meets $f(v)$ in the cap, so $f^*(\varphi)(v) = \varphi(f(v))$: the transpose, the dual map of Lecture 1, with no inner product and no conjugation. The adjoint needs an inner product and conjugates a complex matrix. The inverse transpose is how a group element acts on the contragredient representation, and it needs f invertible. And $f \circ \varphi$ does not compose, since φ lands in the scalars.
2. **Hint.** Draw the cup for $X \otimes Y$ as a cup for Y inside a cup for X, and read off the labels along the top.
   Answer. $Y^* \otimes X^*$
   The nested cup is $1 \to X \otimes Y \otimes Y^* \otimes X^*$: the inner cup for Y sits between the legs of the outer one for X, and each snake for $X \otimes Y$ is a snake for Y inside a snake for X. Putting X^* first would need the strands to cross, which a monoidal category without a braiding cannot draw; even in $\mathrm{Vect}_{\mathrm{fd}}$ it only works after a swap. An object is not in general its own dual. And no braiding is needed, since nesting avoids crossings.
3. **Hint.** Each normalised Bell vector is an unnormalised cup $\sum_i |ii\rangle$ times a number. What does the snake with unnormalised cup and cap give?
   Answer. 0.5
   With $\sum_i |ii\rangle$ as the cup and its transpose as the cap, the snake is straight and gives id. Each of $|\Phi^+\rangle$ and $\langle\Phi^+|$ carries a factor $1/\sqrt{2}$, so $c = \frac{1}{2}$. Counting only one normalisation gives $1/\sqrt{2} \approx 0.707$, and forgetting both gives 1.
4. **Hint.** Which simple object is $s \otimes s \otimes s$? Or use what the theorem on traces says about $d_{X \otimes Y}$.
   Answer. -1
   Here $d_s = -1$, and dimensions are multiplicative, so $d_{s \otimes s \otimes s} = d_s^3 = -1$; equivalently $s \otimes s \otimes s \cong s$. The answer 1 drops the sign of the pivotal structure, as the canonical structure would; 3 adds the factors instead of multiplying them.
5. **Hint.** On which data does $d_V = \mathrm{tr}_L(\mathrm{id}_V)$ depend: the presentation of the dual by a cup and a cap, or the pivotal structure?
   Answer. 2
   The quantum dimension is fixed by the pivotal structure, and for $\mathrm{Vect}_{\mathrm{fd}}$'s a it is the ordinary trace of id_V, so $d_V = 2$ whatever presentation of V^* is used. The loop -2 is the dimension for the other pivotal structure $(-1)^{2j}a$; the ratio -1 is the Frobenius–Schur indicator. The cap is not a free choice: once Ω is the cup, the snake forces $-\Omega^\dagger$. And identifying V^* with V is just a presentation of the dual; d_V is defined for every object.
6. **Hint.** Each of these is self-dual. Find an invariant bilinear form identifying it with its dual, and ask whether the form is symmetric.
   Answer. Spin-$\frac{1}{2}$ of SU(2)
   Spin-$\frac{1}{2}$ is identified with its dual by the antisymmetric Ω, so $j^* = -j$ and $\kappa = -1$. The standard representation of S_3 is real, with a symmetric invariant form; the character formula of the Symmetries course, $\frac{1}{|G|}\sum_g \chi(g^2)$ gives 1. Spin-1 is the real adjoint representation, also with a symmetric form. The sign representation is one-dimensional, and a nonzero form on a line is symmetric.

Lecture 5

1. **Hint.** Apply the truncated Clebsch–Gordan rule: j runs in integer steps from $|j_1 - j_2|$ up to a minimum of two numbers.
   Answer. 1
   The rule runs from $|1-2| = 1$ up to $\min(1+2, k-1-2) = \min(3,1) = 1$, so only $j = 1$ survives. Without the truncation it would be the ordinary $1 \oplus 2 \oplus 3$, but 3 is not even an object of $\mathrm{SU}(2)_4$. Stopping at 2 forgets the second bound $k - j_1 - j_2$. And 0 cannot appear, since $|j_1 - j_2| = 1$.
2. **Hint.** Test each in Vec_G for $G = S_3$, where $g \otimes h = gh$ and $g^* = g^{-1}$.
   Answer. $N_{ab}^{c} = N_{bc^*}^{a^*}$
   Bending wires gives $\mathrm{Hom}(a \otimes b, c) \cong \mathrm{Hom}(b \otimes c^*, a^*)$, so $N_{ab}^{c} = N_{bc^*}^{a^*}$ always; in Vec_G it says $gh = k$ exactly when $hk^{-1} = g^{-1}$. Commutativity fails in Vec_G for S_3, since $gh \neq hg$ in general. The duals reverse the order, $(a \otimes b)^* \cong b^* \otimes a^*$, so the right identity is $N_{ab}^{c} = N_{b^*a^*}^{c^*}$; with a^*b^* it would say $(gh)^{-1} = g^{-1}h^{-1}$, false in S_3. And $N_{ab}^{1} = \delta_{b,a^*}$, which is 0 unless b is the dual of a.
3. **Hint.** The simple objects are 1 and τ. Find d_τ from $\tau \otimes \tau \cong 1 \oplus \tau$ and positivity.
   Answer. 3.618
   With the canonical structure $d_\tau = \mathrm{FPdim}(\tau) = \varphi$, the positive root of $d^2 = 1 + d$. So $\dim C = 1 + \varphi^2 = 2 + \varphi \approx 3.618$. Forgetting the vacuum gives $\varphi^2 \approx 2.618$; adding dimensions instead of their squares gives $1 + \varphi \approx 2.618$ as well; the Yang–Lee root $-1/\varphi$ would give $1 + \varphi^{-2} \approx 1.382$.
4. **Hint.** Count fusion trees: each intermediate label is 1 or τ. When can a 1 follow a 1?
   Answer. 5
   Each intermediate charge is 1 or τ, and two 1's cannot be adjacent, since $1 \otimes \tau \cong \tau$. The counts for $n = 1, 2, 3, 4, 5$ particles with total charge τ obey $f_n = f_{n-1} + f_{n-2}$ and run $1, 1, 2, 3, 5$: so 5. Counting $2^{n-2} = 8$ trees ignores the fusion rules; 3 is the count for four particles.
5. **Hint.** Close the c strand into a loop: the theta graph must be symmetric in a, b and c.
   Answer. $\sqrt{d_a d_b / d_c}\ \mathrm{id}_c$
   The bubble is $\sqrt{d_a d_b / d_c}\ \mathrm{id}_c$; closing it gives the theta graph $\sqrt{d_a d_b / d_c}\, d_c = \sqrt{d_a d_b d_c}$, symmetric as the picture is. The value id_c is Kitaev's unit-norm convention of App. E.1, which differs by $(d_c / d_a d_b)^{1/4}$ per vertex. The factor $\sqrt{d_c/(d_a d_b)}$ is the coefficient in the completeness relation, its inverse. And $d_c\ \mathrm{id}_c$ would close to d_c^2, not symmetric in a, b, c; it also fails the vacuum channel, where $\psi_a^{1a} = \mathrm{id}_a$ has bubble id_a.
6. **Hint.** Which of the two numbers depends on the pivotal structure, and which only on the fusion rules?
   Answer. $\mathrm{FPdim}(s) = 1$ and $d_s = -1$
   The Frobenius–Perron dimension is read off the fusion rules, $s \otimes s \cong 1$: it is the largest eigenvalue of N_s, which is 1, and it is positive whatever the pivotal structure. The quantum dimension is a loop, and with $a_s = -\mathrm{id}$ the loop is -1. The canonical spherical structure would give $d_s = 1 = \mathrm{FPdim}(s)$; this one is not it. And FPdim is never negative.

Lecture 6

1. **Hint.** Compute $L_e^\dagger L_{e'}$ with the bubble twice. What does the answer depend on?
   Answer. The left and right trees are orthogonal bases with one common norm, and the associator is unitary
   By the bubble, $L_e^\dagger L_{e'} = \delta_{ee'}\sqrt{d_a d_b d_c/d_d}\ \mathrm{id}_d$, and the right trees give the same constant; a unitary map between two orthogonal bases of equal norm has a unitary matrix. The pentagon has non-unitary solutions: Yang–Lee, with $[F_\tau^{\tau\tau\tau}]_{11} = -\varphi$, an entry of modulus more than 1. A gauge cannot repair that, since x, A and D are gauge invariant. And Fibonacci's $F_\tau^{\tau\tau\tau}$ has no zero entry, so it is no permutation matrix.
2. **Hint.** First decide whether the left side has trees at all: what must f and l fuse to? Then write the two terms $h = 1, \psi$ of the right side.
   Answer. -1
   The left side is $[F_\psi^{1\sigma\sigma}]_{\sigma 1}[F_\psi^{\sigma\sigma 1}]_{1\sigma}$; its trees would need $f \otimes l = 1 \otimes 1$ to contain ψ, so it is 0. On the right, $h = 1$ gives $\frac{\kappa}{\sqrt{2}} \cdot 1 \cdot \frac{\kappa}{\sqrt{2}}$, since $F_\psi^{\sigma 1\sigma}$ has a unit leg, and $h = \psi$ gives $\frac{\kappa}{\sqrt{2}}\, y\, \frac{\kappa}{\sqrt{2}}$. So $0 = \frac{\kappa^2}{2}(1 + y)$ and $y = -1$, for either sign of κ. Treating the left side as 1 (unit legs) instead of 0 would give $y = +1$, which fails other instances.
3. **Hint.** Compute each entry's factor $u_d^{af}u_f^{bc}/(u_e^{ab}u_d^{ec})$; an entry is invariant when the factor is 1 for every u.
   Answer. $[F_\sigma^{\sigma\sigma\sigma}]_{11}$ in Ising
   For $[F_\sigma^{\sigma\sigma\sigma}]_{11}$, with $e = f = 1$, the factor is $u_\sigma^{\sigma 1}u_1^{\sigma\sigma}/(u_1^{\sigma\sigma}u_\sigma^{1\sigma}) = 1$: it is κ_σ/d_σ. The entry $[F_\sigma^{\sigma\sigma\sigma}]_{1\psi}$ changes under $u_\psi^{\sigma\sigma} \mapsto iu_\psi^{\sigma\sigma}$, Fibonacci's B changes by q^2/p, and $F_\psi^{\sigma\psi\sigma}$ changes under $u_\sigma^{\psi\sigma} \mapsto -u_\sigma^{\psi\sigma}$.
4. **Hint.** Two pentagon instances give $A^2 + A = 1$; which root can sit in a unitary matrix?
   Answer. 0.6180339887
   The pentagon forces $A^2 + A = 1$, with roots $\varphi^{-1} \approx 0.618$ and $-\varphi \approx -1.618$. A unitary matrix has entries of modulus at most 1, so $A = \varphi^{-1}$; this is also κ_τ/d_τ with $d_\tau = \varphi$. The root $-\varphi$ is Yang–Lee; $\varphi^{-1/2} \approx 0.786$ is the off-diagonal entry in the symmetric gauge.
5. **Hint.** Which entries are gauge invariant, and how many values can they take?
   Answer. 2
   Every solution has $x = 1$, $D = -A$, $BC = A$ and $A \in \{\varphi^{-1}, -\varphi\}$, and a gauge with $u \neq 0$ can make $B = C$ with either sign: two classes, Fibonacci and Yang–Lee, as Ocneanu rigidity allows. Counting the sign of B gives 4, but that sign is a gauge; counting only unitary solutions gives 1.
6. **Hint.** Find a number built from F that no gauge, and no monoidal equivalence, can change.
   Answer. They are inequivalent: $\kappa = d_\sigma[F_\sigma^{\sigma\sigma\sigma}]_{11} = \pm 1$ is a gauge invariant
   The invariant $\kappa = d_\sigma[F_\sigma^{\sigma\sigma\sigma}]_{11}$ is $+1$ for H and -1 for $-H$, and an equivalence acts on F-symbols as a gauge transformation, so the categories differ: Ising and the category of $\mathrm{SU}(2)_2$. As quantum gates H and $-H$ agree up to a phase, but an F-symbol is not a gate. The gauge $u_1^{\sigma\sigma} = -1$ multiplies $[F_\sigma^{\sigma\sigma\sigma}]_{11}$ by $u_\sigma^{\sigma 1}u_1^{\sigma\sigma}/(u_1^{\sigma\sigma}u_\sigma^{1\sigma}) = 1$, so it cannot. The fusion rules are the same.

Lecture 7

1. **Hint.** Draw σ_i twice in the plane. Can the double exchange be undone without passing a strand through another?
   Answer. $\sigma_i^2 = 1$
   Twice the same exchange in a plane takes one particle once around the other, a loop that cannot be shrunk; in three dimensions it can, which is why S_n has $\sigma_i^2 = 1$. The braid relation (Reidemeister III) and far commutation (strands that do not meet) hold in both groups. So B_n is not S_n: adding $\sigma_i^2 = 1$ to B_n gives S_n.
2. **Hint.** Try a diagonal R, say $\mathrm{diag}(1, 1, 1, 2)$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$: it satisfies $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$. Test each candidate c in the braid form.
   Answer. $c = P \circ R$, with P the flip
   The braid form needs the factors to move the strands, which the flip does: $c = P \circ R$, Kassel's $\tau \circ R$. R itself, its inverse and its transpose act on fixed positions: for $R = \mathrm{diag}(1, 1, 1, 2)$, which satisfies the first equation, each of them fails the braid form, while $P \circ R$ satisfies it. Three different things are called R: Integrability's $R(\lambda)$, the braiding $c = P \circ R$, and the R-symbols.
3. **Hint.** The quadratic has two roots on the unit circle, with product 1; find them from their sum. Which one squares to R_1?
   Answer. 0.7
   The roots of $x^2 + \varphi^{-1}x + 1 = 0$ have product 1 and sum $-\varphi^{-1} = 2\cos(3\pi/5)$, so they are $e^{\pm 3\pi i/5}$. Squaring, $e^{-6\pi i/5} = e^{4\pi i/5} = R_1$, while $e^{6\pi i/5} \neq R_1$: so $R_\tau = e^{-3\pi i/5} = e^{7\pi i/5}$ and $s = 0.7$. The root $s = 0.3$ is the course's own chirality, with $R_1 = e^{-4\pi i/5}$; $s = 0.2$ squares to R_1 but does not solve the quadratic.
4. **Hint.** Use $\theta_a = d_a^{-1} \sum_c N_{aa}^c d_c R_c^{aa}$, and $\varphi^2 = \varphi + 1$.
   Answer. 0.4
   $\theta_\tau = (e^{-4\pi i/5} + \varphi\, e^{3\pi i/5})/\varphi = e^{4\pi i/5}$, so $s = 2/5 = 0.4$. Averaging the two R-symbols without the dimensions gives a number of modulus less than 1; the conjugate chirality gives $s = 0.6$; reading off $R_\tau^{\tau\tau}$ alone gives 0.3.
5. **Hint.** A braiding is one number $c = c_{s,s}$. Which equation does the hexagon impose on it?
   Answer. 4
   The hexagon forces $c^2 = \omega(s, s, s)$: for $\omega = 1$, $c = \pm 1$ (boson and fermion); for $\omega = -1$, $c = \pm i$ (semion and antisemion). That is 4. Counting one per category gives 2; allowing every fourth root of unity on each gives 8.
6. **Hint.** A gauge changes R_c^{ab} by u_c^{ba}/u_c^{ab}. Which combination cancels the factor?
   Answer. The double braiding $R_c^{ab}R_c^{ba}$
   R_c^{ab} changes by u_c^{ba}/u_c^{ab} and R_c^{ba} by the inverse, so their product, the double braiding, is invariant; for e and m in the toric code it is -1. For $a \neq b$ with neither label 1, R_c^{ab} alone changes (choose $u_c^{ab} = i$, $u_c^{ba} = 1$), and so does its square, by $(u_c^{ba}/u_c^{ab})^2$; with a unit leg the vertices are fixed and $R = 1$. The F-symbol $[F_\sigma^{\sigma\sigma\sigma}]_{1\psi}$ changes under $u_\psi^{\sigma\sigma} \mapsto i u_\psi^{\sigma\sigma}$ (Lecture 6).

Lecture 8

1. **Hint.** Δ must be an algebra map, so it must preserve brackets; and some counit must satisfy $(\varepsilon_H \otimes \mathrm{id})\Delta = \mathrm{id} = (\mathrm{id} \otimes \varepsilon_H)\Delta$.
   Answer. $\Delta(x) = x \otimes 1 + 1 \otimes x$
   The primitive coproduct preserves brackets, $[\Delta x, \Delta y] = [x, y] \otimes 1 + 1 \otimes [x, y]$, and $\varepsilon_H(x) = 0$ is a counit: this is the total angular momentum of QM-I. $x \otimes x$ breaks brackets: $[e \otimes e, f \otimes f] \neq h \otimes h$ on $\mathbb{C}^2 \otimes \mathbb{C}^2$. $x \otimes 1$ preserves brackets, but $\mathbb{C}$ is not a unit: x acts on $\mathbb{C} \otimes V$ by the scalar $\varepsilon_H(x)$ and on V by $\rho(x)$, so the unitor $\mathbb{C} \otimes V \to V$ is not a module map, whatever $\varepsilon_H(x)$ is. The factor $\frac{1}{2}$ breaks brackets, since $[\Delta e, \Delta f]$ then carries $\frac{1}{4}$.
2. **Hint.** Expand f in the dual basis and compute $(x \cdot f)(e_j)$.
   Answer. $-\rho(x)^T$
   $(x \cdot f)(e_j) = -f(\rho(x)e_j) = -\sum_i f_i \rho(x)_{ij}$, so the column of coefficients transforms by $-\rho(x)^T$; then the evaluation is a module map, $\mathrm{ev}(\Delta(x)(f \otimes v)) = -f(xv) + f(xv) = 0 = \varepsilon_H(x) f(v)$, and the weights of V^* are opposite. With $\rho(x)^T$, the antipode replaced by the identity, the evaluation is not a module map. $-\rho(x)$, acting on columns, is not even a representation: $[-\rho(x), -\rho(y)] = \rho([x, y])$. $\rho(x)^{-1}$ confuses the Lie algebra with the group, where $S_H(g) = g^{-1}$; and $\rho(e)$ is not invertible.
3. **Hint.** Write $c = P \circ R$ on the basis $|ab\rangle$ of $W \otimes W$. Which basis vectors does it send to multiples of themselves?
   Answer. 0
   $c\,|ab\rangle = (-1)^{ab}\,|ba\rangle$. Only $|00\rangle$ and $|11\rangle$ are sent to multiples of themselves, with +1 and −1; $|01\rangle$ and $|10\rangle$ are exchanged and contribute nothing. So $\mathrm{tr}\, c = 0$. The trace of R alone is $1 + 1 + 1 - 1 = 2$, and so is the trace of the flip P: confusing c with R, or forgetting R, gives 2. This R makes the odd line a fermion, $c_{V_1,V_1} = -1$: the braiding of sVec.
4. **Hint.** A fibre functor preserves Frobenius–Perron dimensions, and in Vect they are integers. Is that condition enough?
   Answer. Vec_G for $G = \mathbb{Z}_3$, whatever its pivotal structure
   Vec_G has the fibre functor that forgets the grading, so by reconstruction it is $\mathrm{Rep}(H)$, with $H = \mathbb{C}^G$; a pivotal structure with $d = e^{2\pi i/3}$ does not matter, because the obstruction is FPdim, not d. Fibonacci ($\mathrm{FPdim}\,\tau = \varphi$) and Ising ($\mathrm{FPdim}\,\sigma = \sqrt{2}$) have none. Vec_G^ω with ω not a coboundary has every FPdim 1 but no fibre functor, which would trivialise ω: integer FPdims are necessary, not sufficient.
5. **Hint.** How many dimensions does the triplet have? Check your answer against the matrix of B_q.
   Answer. -4
   $\det B_q = q^3 \cdot (-q^{-1}) = -q^2 = -4$. The matrix agrees: $q \cdot q \cdot \det \begin{pmatrix} q - q^{-1} & 1 \\ 1 & 0 \end{pmatrix} = -q^2$.
   Counting the triplet once gives −1; dropping the sign gives 4; omitting the singlet gives $q^3 = 8$.
6. **Hint.** $c_1 c_2 = \lambda^2 B_1 B_2$. What must λ^2 be?
   Answer. $\lambda = A^{-1}$, for a root A of $q = A^2$
   $\lambda^2 q = 1$, so $\lambda = \pm q^{-1/2}$, which is A^{-1} for one of the two roots A of q; then $c = A^{-1} B_q$ has eigenvalues A and $-A^{-3}$. $\lambda = 1$ leaves the factor q; $\lambda = q^{-1}$ leaves q^{-1}; $\lambda = A$ gives q^2.

Lecture 9

1. **Hint.** Turn a belt that holds two strands through a full turn. What happens to each strand, and to the two strands relative to each other?
   Answer. $\theta_{X\otimes Y} = c_{Y,X}\,c_{X,Y}\,(\theta_X \otimes \theta_Y)$
   A full turn of the belt twists each strand and wraps the two once around each other: the double braiding $c_{Y,X}c_{X,Y}$ times $\theta_X \otimes \theta_Y$. On a channel c this is the ribbon relation $R_c^{ab}R_c^{ba} = \theta_c/(\theta_a\theta_b)$. Without the double braiding the toric code fails: $\theta_e = \theta_m = 1$ but $\theta_{e\otimes m} = \theta_\varepsilon = -1$, the double braiding of e and m. A single crossing does not even map $X \otimes Y$ to itself. The inverse double braiding fails in Fibonacci: on the channel $\tau \otimes \tau \to 1$ the double braiding is $(R_1^{\tau\tau})^2 = e^{-8\pi i/5} = \theta_\tau^{-2}$, from Lecture 7's $R_1^{\tau\tau} = e^{-4\pi i/5}$ and $\theta_\tau = e^{4\pi i/5}$, so it would give $\theta_\tau^2\theta_\tau^2 = \theta_\tau^4 \neq 1 = \theta_1$.
2. **Hint.** Expand the crossing of the curl: one term closes a loop, the other straightens the strand.
   Answer. 0.75
   The curl is $A\delta + A^{-1} = -A^3$, here $-e^{i\pi/2} = -i = e^{3\pi i/2}$, so $s = 0.75$. This is $\theta_{1/2}$ of $\mathrm{SU}(2)_1$ in the Temperley–Lieb pivotal structure; the pivotal flip turns it into $+i$, $s = 0.25$, the semion's spin, which is also what forgetting the minus sign or taking the opposite curl $-A^{-3}$ gives.
3. **Hint.** Multiply a crossing by its inverse, $(A + A^{-1}U)(A^{-1} + AU)$, and use $U^2 = \delta U$.
   Answer. $\delta = -A^2 - A^{-2}$
   $(A + A^{-1}U)(A^{-1} + AU) = 1 + (A^2 + A^{-2})U + U^2 = 1 + (A^2 + A^{-2} + \delta)U$, which is 1 for every A exactly when $\delta = -A^2 - A^{-2}$. The value -2 is this at $A = \pm 1$ only, the Penrose loop of Lecture 4; $+A^2 + A^{-2}$ doubles the U term instead of cancelling it; and $A^4 = 1$ is not needed.
4. **Hint.** Closing the identity on two strands gives two loops; closing U gives one. Then use $\delta = -A^2 - A^{-2}$.
   Answer. 2
   The trace is $\delta^2 - \delta/\delta = \delta^2 - 1$, and $\delta = -2\cos(\pi/6) = -\sqrt{3}$, so it is $2 = [3]_q$ at $q = e^{i\pi/6}$: the quantum dimension of spin 1 in $\mathrm{SU}(2)_4$, deformed from the classical 3. Forgetting the -1 gives 3; $-\delta = \sqrt{3} \approx 1.732$ is the dimension of spin $\frac{1}{2}$ there.
5. **Hint.** Which quantum integers vanish at this A, and what does the recursion for p_{k+1} need?
   Answer. It is negligible: its trace $(-1)^{k+1}[k+2]_q$ vanishes, and it closes to zero in every network
   At this A, $[k+2]_q = 0$ while $[m]_q \neq 0$ for $m \leq k+1$, so the recursion reaches p_{k+1}, an idempotent like the others; its trace $(-1)^{k+1}[k+2]_q$ is zero, and by Kauffman–Lins it closes to zero in every network, so it is negligible and the quotient sets it to zero. The recursion fails only at p_{k+2}, and the twist of p_{k+1} is not 1.
6. **Hint.** Compute θ_1 in the Temperley–Lieb structure first. What does the flip by $(-1)^n$ do to it?
   Answer. 0.1875
   $\theta_1 = -A^3 = -e^{3\pi i/8}$, and the flip multiplies it by $(-1)^1$: $\theta_{1/2} = e^{3\pi i/8}$, $s = 3/16 = 0.1875$, the value $e^{2\pi ij(j+1)/(k+2)}$ of Lecture 9 and Kitaev's $\nu = 3$. Ising has $\theta_\sigma = e^{i\pi/8}$, $s = 1/16 = 0.0625$: same fusion rules, different category ($\kappa = +1$ against -1). Forgetting the flip leaves $-e^{3\pi i/8}$, $s = 0.6875$.

Lecture 10

1. **Hint.** For an abelian model each fusion has one channel, so $\tilde{S}_{ab}$ is a phase. Which phase does e pick up around m?
   Answer. -0.5
   $\tilde{S}_{em} = \sum_c N_{e\bar{m}}^c d_c \theta_c/(\theta_e\theta_m) = \theta_\varepsilon = -1$, the phase of taking e around m, and $S = \tilde{S}/\sqrt{4}$ gives $-\frac{1}{2}$. Forgetting the normalisation gives -1; $+\frac{1}{2}$ is S_{ee}.
2. **Hint.** S is invertible exactly when the Müger centre is trivial. Which of these has a transparent object other than 1?
   Answer. Fibonacci
   In a symmetric category every object is transparent, so sVec, $\mathrm{Rep}(S_3)$ and the bosonic $\mathbb{Z}_2$ all have singular S (for sVec, $S = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$). In Fibonacci only 1 is transparent, and $S = (2+\varphi)^{-1/2}\left(\begin{smallmatrix}1&\varphi\\\varphi&-1\end{smallmatrix}\right)$ has determinant -1.
3. **Hint.** Compute $N_{\sigma\sigma}^c = \sum_x S_{\sigma x} S_{\sigma x} \overline{S_{cx}}/S_{1x}$ for each c; the column $x = \sigma$ drops out.
   Answer. $1 \oplus \psi$
   For $c = 1$: $\frac{1}{2} + 0 + \frac{1}{2} = 1$; for $c = \sigma$: $\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 0$; for $c = \psi$: $\frac{1}{2} + \frac{1}{2} = 1$. So $\sigma \otimes \sigma = 1 \oplus \psi$, the fusion rule of Lecture 5, recovered from S alone.
4. **Hint.** $\sum_a d_a^2 \theta_a = \sqrt{\dim C}\, e^{2\pi i c/8}$.
   Answer. 1
   $1 + i = \sqrt{2}\, e^{i\pi/4}$, so $2\pi c/8 = \pi/4$ and $c = 1$. Reading the phase of θ_s alone gives $c = 2$; the antisemion, $\theta = -i$, has $c = -1 = 7$.
5. **Hint.** Close the strand: the loop becomes the Hopf link $\tilde{S}_{\tau\tau}$. What must λ be multiplied by to give it?
   Answer. -0.618034
   Closing the strand gives $\lambda d_\tau = \tilde{S}_{\tau\tau}$, and $\tilde{S}_{\tau\tau} = \theta_\tau^{-2} + \varphi\,\theta_\tau^{-1} = -1$ (Lecture 10), so $\lambda = -1/\varphi = -\varphi^{-1} \approx -0.618$. Forgetting to divide by d_τ gives -1; dividing by $\sqrt{\dim C}$ instead gives the entry $S_{\tau\tau} \approx -0.526$. This is the value at τ of the character λ_τ of Lecture 10's lemma that loops fuse: $\lambda_\tau(\tau)^2 = 1 + \lambda_\tau(\tau)$, as $\tau \otimes \tau = 1 \oplus \tau$ requires.
6. **Hint.** If the loop is nonzero on x, the proof shows $\lambda_x(b) = d_b$ for every b. What does that say about the double braiding of x with everything?
   Answer. Only 1 is transparent, and every $d_c > 0$ and $|\theta_c| = 1$ (unitarity)
   If $\Omega_x \neq 0$ then $\sum_c N_{x\bar{b}}^c d_c\, \theta_c/(\theta_x\theta_b) = \sum_c N_{x\bar{b}}^c d_c$; with $d_c > 0$ and phases of modulus 1, every phase is 1, so x is transparent, and a trivial Müger centre forces $x = 1$. Integer dimensions and pointedness are not used (Fibonacci has neither), and the symmetry of S holds in every premodular category.

Lecture 11

1. **Hint.** Compose $\mu \circ (\mu \otimes \mathrm{id})$ and $\mu \circ (\mathrm{id} \otimes \mu) \circ \alpha_{g,h,k}$ on $(\delta_g \otimes \delta_h) \otimes \delta_k$ and read off the scalars.
   Answer. $\psi(g,h)\,\psi(gh,k) = \omega(g,h,k)\,\psi(h,k)\,\psi(g,hk)$
   Multiplying g, h first gives $\psi(g,h)\psi(gh,k)$; reassociating first costs $\omega(g,h,k)$ and then gives $\psi(h,k)\psi(g,hk)$. So $\omega|_H = (\delta\psi)^{-1}$, the rank-one module pentagon in Lecture 11. Without ω it is the equation for $\omega = 1$, a 2-cocycle; with ω on the other side it is EGNO's printed $\omega = d\psi$, which their own conventions contradict; $\psi(g,h)\psi(h,k)$ pairs the wrong products.
2. **Hint.** They are pairs (H, ψ) with $\omega|_H$ a coboundary. Which subgroups H qualify?
   Answer. 1
   For $H = 1$ the condition is empty: the regular module. For $H = \mathbb{Z}_2$ the module pentagon at (s,s,s) reads $\psi(s,s) = -\psi(s,s)$, impossible, so the rank-one module $\mathrm{Vect}_{\mathrm{fd}}$ does not exist (and there is no fibre functor, Lecture 8). That leaves 1. Without the twist the answer would be 2.
3. **Hint.** Count the pairs $(H, [\psi])$: list the subgroups of $\mathbb{Z}_4$, and the classes $[\psi]$ on each.
   Answer. 3
   The subgroups are 1, $\mathbb{Z}_2 = \{0, 2\}$ and $\mathbb{Z}_4$; each is cyclic, so $H^2(H, \mathbb{C}^\times) = 0$ and each gives one module category, $\mathrm{Mod}(\mathbb{C}[H])$, with 4, 2 and 1 simple objects. The group is abelian, so conjugation identifies nothing: 3. Counting elements gives 4; counting only the proper subgroups gives 2.
4. **Hint.** Use $[M,N] = \bigoplus_c \mathrm{Hom}(c \otimes M, N) \otimes c$: for which c is $c \otimes \delta_x = \delta_y$?
   Answer. $\delta_{yx^{-1}}$
   $\delta_c \otimes \delta_x = \delta_{cx}$ equals δ_y exactly for $c = yx^{-1}$, so $[\delta_x, \delta_y] = \delta_{yx^{-1}}$; equivalently $[x,y] = y \otimes x^*$ by duality. In S_3 the order matters: with $x = (1\,2)$ and $y = (2\,3)$, yx^{-1} differs from $x^{-1}y$ and from xy^{-1}; with $x = (1\,2\,3)$ and $y = (1\,2)$, it differs from $y^{-1}x$.
5. **Hint.** Test each statement on Vec_G^ω with ω not a coboundary, and on the module $\mathrm{Vect}_{\mathrm{fd}}$ of Vec_G.
   Answer. Every indecomposable semisimple module category over a fusion category C is equivalent to $\mathrm{Mod}_C(A)$ for an algebra A in C
   Ostrik's theorem: $A = [M,M]$ for any nonzero M works, and the algebra lives in C. The regular module is not the only one: $\mathrm{Vect}_{\mathrm{fd}}$ over Vec_G is another, with one simple object instead of $|G|$. $\mathrm{Vect}_{\mathrm{fd}}$ is a rank-one module only when ω is a coboundary (Lecture 11; the semion's category has none). And $[M,M]$ need not be 1: it is 1 for an invertible object of the regular module, $[\delta_x, \delta_x] = \delta_1$, but for $\mathrm{Vect}_{\mathrm{fd}}$ over Vec_G it is $\bigoplus_g \delta_g$.
6. **Hint.** The simple objects are free modules $\delta_g \otimes A$. When are two of them isomorphic?
   Answer. 2
   $\mathrm{Hom}_A(\delta_g \otimes A, \delta_{g'} \otimes A) \cong \mathrm{Hom}(\delta_g, \delta_{g'} \otimes A)$ is nonzero exactly when $g \in g' + H$, so the simple objects are the cosets G/H: $6/3 = 2$. Counting the elements of H gives 3, and counting all of G gives 6.

Lecture 12

1. **Hint.** A module endofunctor of a left module must commute with the left action. Then compose two of the candidates and compare with gh.
   Answer. $g \mapsto (- \otimes \delta_{g^{-1}})$
   Tensoring on the left by δ_g does not commute with the left action of Vec_G unless g is central, so it is not a module functor at all. Tensoring on the right does, and $(- \otimes \delta_{g^{-1}}) \circ (- \otimes \delta_{h^{-1}}) = - \otimes \delta_{h^{-1}g^{-1}} = - \otimes \delta_{(gh)^{-1}}$, which respects the product. With δ_g in place of $\delta_{g^{-1}}$ the composite is $- \otimes \delta_{hg}$: the order is reversed, which is why the regular module's dual is $C^{\otimes \mathrm{op}}$ in general.
2. **Hint.** The braiding $c_{X,Y}$ of the centre is the half-braiding of one of the two objects, evaluated on the other. Which one, and on what?
   Answer. $c_{e,m} = 1$ and $c_{m,e} = -1$
   The braiding is $c_{(Z,\gamma),(Z',\gamma')} = \gamma'_Z$: the half-braiding of the second object, evaluated on the first. So $c_{e,m}$ is m's half-braiding on δ_0, $\chi_m(0) = 1$, since the character of m is trivial; and $c_{m,e}$ is e's half-braiding on δ_1, $\chi_-(1) = -1$. Using the first object's half-braiding swaps the two values. The braiding is not symmetric. A single entry can change with the gauge, but the double braiding $c_{m,e}c_{e,m} = -1$ cannot: it is the phase of taking e around m in the toric code. Both entries -1 would make the double braiding $+1$; both $+1$ would make e and m transparent to each other.
3. **Hint.** A half-braiding needs an isomorphism $\gamma_{\delta_h} : \delta_h \otimes \delta_g \to \delta_g \otimes \delta_h$ for every h. Where do the two sides live?
   Answer. 0
   $\delta_h \otimes \delta_g = \delta_{hg}$ and $\delta_g \otimes \delta_h = \delta_{gh}$, and morphisms of Vec_{S_3} preserve the grading, so γ_{δ_h} is zero unless $hg = gh$. A transposition does not commute with a 3-cycle, so no isomorphism exists: 0. A transposition can enter the centre only inside the sum over its whole conjugacy class, with a representation of its centraliser $\mathbb{Z}_2$; counting those two representations gives 2.
4. **Hint.** The labels are pairs of a conjugacy class and an irreducible representation of the centraliser of one of its elements. Find each centraliser, and count its irreducible representations.
   Answer. 22
   The centralisers are D_4, D_4, $\mathbb{Z}_4$, $\{1, r^2, s, sr^2\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$ and $\{1, r^2, sr, sr^3\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, with 5, 5, 4, 4, 4 irreducible representations: 22. Pairs of an element and an irrep of G give $8 \times 5 = 40$, and pairs of a class and an irrep of G give 25; both are the trap of using G in place of the centraliser.
5. **Hint.** Müger's theorem gives the dimension of a centre in terms of $\dim C$.
   Answer. 13.09
   $\dim \mathcal{Z}(C) = (\dim C)^2$, and $\dim \mathrm{Fib} = 1 + \varphi^2 = 2 + \varphi \approx 3.618$, so $\dim \mathcal{Z}(\mathrm{Fib}) = (2 + \varphi)^2 \approx 13.090$. It agrees with $\mathrm{Fib} \boxtimes \mathrm{Fib}^{\mathrm{rev}}$, whose four simple objects have dimensions $1, \varphi, \varphi, \varphi^2$. Forgetting the square gives 3.618.
6. **Hint.** Use the theorem of Etingof, Nikshych and Ostrik: compare the centres, or at least their dimensions and spins.
   Answer. Vec_G and $\mathrm{Rep}(G)$ are Morita equivalent for every finite group G
   The module category $\mathrm{Vect}_{\mathrm{fd}}$ of Vec_G has dual $\mathrm{Rep}(G)$, and $\mathrm{Rep}(G)^{\otimes \mathrm{op}} \simeq \mathrm{Rep}(G)$ by the flip, so they are Morita equivalent; for $G = S_3$ they are not tensor equivalent, since $\mathrm{Rep}(S_3)$ has a simple object of dimension 2. The twisted $\mathrm{Vec}_{\mathbb{Z}_2}$ has the double semion as centre, with spins $\pm i$, which the toric code lacks. And $\dim \mathcal{Z}(\mathrm{Fib}) \approx 13.09$ differs from $\dim \mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2}) = 4$.

Lecture 13

1. **Hint.** A Lagrangian algebra here is connected, étale and of dimension 2. On $X \otimes X \cong 1$ the braiding $c_{X,X}$ acts by θ_X: what does commutativity, $\mu \circ c = \mu$, ask of it?
   Answer. $X = \varepsilon$
   With $\lambda \neq 0$ the multiplication $X \otimes X \to 1$ is non-zero (separability needs it), and $\mu \circ c_{X,X} = \mu$ forces $\theta_X = 1$. So $1 \oplus e$ and $1 \oplus m$ are Lagrangian ($\theta = 1$), and $1 \oplus \varepsilon$ is not ($\theta_\varepsilon = -1$). The square-zero algebra on $1 \oplus \varepsilon$ is commutative, but it is not separable, so it is not étale.
2. **Hint.** By the theorem of Davydov, Müger, Nikshych and Ostrik, count the indecomposable module categories over $\mathrm{Vec}_{\mathbb{Z}_2 \times \mathbb{Z}_2}$ instead: pairs $(H, [\psi])$.
   Answer. 6
   Lagrangian algebras in $\mathcal{Z}(C)$ correspond to indecomposable C-module categories, and over Vec_G these are pairs $(H, [\psi])$ with $[\psi] \in H^2(H, \mathbb{C}^\times)$, up to conjugating both, which does nothing for an abelian group (Lecture 11). The subgroups are 1, three copies of $\mathbb{Z}_2$ and $\mathbb{Z}_2 \times \mathbb{Z}_2$; the first four carry one class each and the last two, so $1 + 3 + 2 = 6$. Forgetting the second class gives 5, one for each subgroup.
3. **Hint.** An X on an edge flips the plaquettes containing it. How many plaquettes contain an edge of the boundary row?
   Answer. m
   A smooth-boundary edge lies in one plaquette, so an X-string on the dual lattice that ends by crossing it violates only the plaquette at its other end: m is absorbed. Every edge there still has two ends with stars, so near that boundary every operator flips an even number of stars and a single e survives; $\varepsilon = e \otimes m$ becomes e. At the rough boundary it is the other way round: e condenses.
4. **Hint.** Apply the two assignments $X_j \mapsto Z_{j-1}Z_j$ and $Z_{j-1}Z_j \mapsto X_{j-1}$ in turn. Where does X_j end up?
   Answer. $D^2 = (1 + \eta)\,\mathcal{T}^{-1}$
   Twice the map sends X_j to X_{j-1}: a translation by one site backwards, $\mathcal{T}^{-1}$. With the projection onto η-even states this gives $D^2 = (1 + \eta)\mathcal{T}^{-1}$ (Seiberg, Seifnashri and Shao). The continuum rule $N \otimes N = 1 \oplus \eta$ drops the translation; $D^2 = 1$ would make D invertible, but $D\eta = D$ kills the η-odd states. $\mathcal{T}$ in place of $\mathcal{T}^{-1}$ is the square of Shao's operator, $D_{\mathrm{Shao}} = D^\dagger/\sqrt{2}$, up to the factor $\frac{1}{2}$.
5. **Hint.** The three invertible objects have dimension 1. Read d_N off the fusion rule $N \otimes N = \bigoplus_{a \in A} a$.
   Answer. 6
   $d_N^2 = d_{N \otimes N} = |A| = 3$, so $d_N = \sqrt{3}$, and the global dimension is $3 \cdot 1 + 3 = 6 = 2\,|A|$. Taking $d_N = 1$ gives 4; squaring the number of simple objects gives 16. For $A = \mathbb{Z}_2$ this is the Ising category's 4.
6. **Hint.** A rank-one module over Vec_G^ω is a ψ solving the module pentagon with $H = G$. When does one exist?
   Answer. Vec_G^ω for $G = \mathbb{Z}_2$ and $\omega(s, s, s) = -1$
   A rank-one module exists exactly when $[\omega] = 0$. For the semion's ω the pentagon at (s, s, s) reads $\psi(s, s) = -\psi(s, s)$, so there is none: the symmetry is anomalous, although its regular module category exists. The untwisted Vec_G always has $\mathrm{Vect}_{\mathrm{fd}}$, and full gauging gives the dual symmetry $\mathrm{Rep}(G)$; for $\mathbb{Z}_2 \times \mathbb{Z}_2$ there are even two choices of ψ.

Lecture 14

1. **Hint.** Which bordism compares μ with μ composed with an exchange of its two inputs, and what must Z do with that exchange?
   Answer. The swap of the two legs of the pair of pants is a bordism, and a symmetric monoidal Z sends it to the flip of Vect
   Composing the pants with the swap of its legs gives a diffeomorphic bordism, so $\mu \circ Z(\text{swap}) = \mu$; a symmetric monoidal functor sends the swap to the flip P of Vect, so $\mu \circ P = \mu$. Matrix algebras with the trace are Frobenius and not commutative. A drawing is not an argument. And the issue is not the target: Kock's counterexample keeps Vect and drops only the symmetry of Z, which then gives graded-commutative algebras.
2. **Hint.** Find $\Delta(e_i)$ from the pairing, then the handle element $w = \mu \circ \Delta \circ \iota$, and use $Z(\Sigma_g) = \varepsilon_A(w^g)$.
   Answer. 5
   The pairing $\varepsilon_A(e_i e_j) = \delta_{ij}\lambda_i$ gives $\Delta(e_i) = \lambda_i^{-1} e_i \otimes e_i$, so $w = \sum_i \lambda_i^{-1} e_i$ and $Z(\Sigma_g) = \sum_i \lambda_i^{1-g}$. At $g = 3$: $1^{-2} + (\frac{1}{2})^{-2} = 1 + 4 = 5$. With the exponent $g - 1$ in place of $1 - g$ one gets 1.25; $Z(S^2) = 1.5$ and $Z(T^2) = 2$ are the genus-0 and genus-1 values.
3. **Hint.** Use Mednykh's formula: a sum over the irreducible characters of a power of $|G|/\dim\chi$.
   Answer. 272
   $Z(\Sigma_g) = \sum_\chi (|G|/\dim\chi)^{2g-2}$, so at $g = 2$: $4 \cdot 8^2 + 4^2 = 256 + 16 = 272$, and $\#\,\mathrm{Hom}(\pi_1\Sigma_2, D_4) = 272 \cdot 8 = 2176$. Forgetting the division by $|G|$ gives 2176; the exponent with $\dim\chi/|G|$ gives 0.125. Rescaling the counit by λ would multiply the answer by λ^{-1}, which is why the normalisation $\varepsilon_A(f) = f(1)/|G|$ is fixed.
4. **Hint.** Atiyah's axioms give $Z(\Sigma \times S^1)$ in terms of $Z(\Sigma)$. Then count the states on the torus: $Z_{\mathrm{TV},C} = \mathrm{RT}_{\mathcal{Z}(C)}$.
   Answer. 8
   $Z(\Sigma \times S^1) = \dim Z(\Sigma)$, the trace of the identity. The torus carries rank $\mathcal{Z}(C)$ states, and $\mathcal{Z}(\mathrm{Vec}_{S_3})$ has $3+2+3 = 8$ simple objects, so the answer is 8. The 2D Dijkgraaf–Witten value $Z(T^2) = 3$, the number of conjugacy classes, belongs to a different theory; 64 squares the dimension; 18 is the trap of pairs (g, irrep of G).
5. **Hint.** For unitary C, $p_- = \overline{p_+}$ and $p_+/p_- = e^{2\pi i c/4}$. The anomaly vanishes when $p_+ = p_-$.
   Answer. The toric code $\mathcal{Z}(\mathrm{Vec}_{\mathbb{Z}_2})$, with $c = 0$
   The toric code has four simple objects of dimension 1 with $\theta = 1, 1, 1, -1$, so $p_+ = p_- = 2$: no anomaly. It is a centre, and every centre has $c = 0$ (Lecture 12), which is why $\mathrm{RT}_{\mathcal{Z}(C)}$ can equal the Turaev–Viro theory of C. Fibonacci ($p_+/p_- = e^{2\pi i \cdot 7/10}$), the semion ($p_+/p_- = i$) and Ising ($p_+/p_- = e^{i\pi/4}$) are anomalous, so none of them is a centre.
6. **Hint.** Recall the proof sketch: which equation makes the weights invariant under the 2–3 Pachner move, and what does the vertex factor need?
   Answer. A spherical fusion category with $\dim C \neq 0$; no braiding is used
   The tetrahedra are weighted by F-symbols, and invariance under the 2–3 move is the pentagon; the edges are weighted by dimensions, which need a spherical structure, and the vertex factor $(\dim C)^{-1}$ needs $\dim C \neq 0$. No braiding enters: a modular category such as Fibonacci is accepted, but its braiding is ignored, and $Z_{\mathrm{TV,Fib}} = \mathrm{RT}_{\mathrm{Fib}\boxtimes\mathrm{Fib}^{\mathrm{rev}}}$. Modularity is what Reshetikhin–Turaev needs. Turaev–Viro never yields Fibonacci alone because it yields $\mathrm{RT}_{\mathcal{Z}(C)}$, and Fibonacci, with $c = \frac{14}{5}$, is not a centre.